

Manual 48 — Singapore AI Governance: MGF, GenAI, AI Verify & Agentic AI

****Status:**** CONTROLLED DEVELOPMENT

****Version:**** 0.3.0-draft

****Source verification date:**** 1 September 2026

****Canonical stage:**** 3 — controlled English training package

Purpose

Operationalize Singapore's practical AI-governance ecosystem for enterprise use, emphasizing governance design, assurance, testing, generative AI and agentic AI. Manual 48 remains one canonical stage behind Manual 47 and must not overtake it.

Controlled Stage-3 package

The branch now contains:

- refreshed authoritative-source register;
- controlled English training modules;
- practical scenarios;
- control/evidence workbook; and
- fail-closed preflight.

Architecture

1. Singapore AI-governance ecosystem and regulatory context.
2. Model AI Governance Framework foundations.
3. Internal governance structures and accountability.
4. Human involvement in AI-augmented decision-making.
5. Operations management across the AI lifecycle.
6. Stakeholder interaction and communication.
7. Model AI Governance Framework for Generative AI.
8. GenAI ecosystem dimensions and shared responsibility.
9. AI Verify governance principles and testing approach.
10. Evidence, assurance and testability.
11. Model AI Governance Framework for Agentic AI.
12. Assessing and bounding agent capabilities and risks.
13. Human accountability and meaningful approval checkpoints.
14. Lifecycle technical controls and process controls.
15. End-user transparency, responsibility and education.
16. May 2026 case-study and best-practice updates.
17. Mapping Singapore governance to Manual 46 universal controls.
18. Mapping to EU AI Act, ISO/IEC 42001 and NIST AI RMF without false equivalence.
19. Enterprise implementation blueprint.
20. Scenario-based training exercises.
21. Control and evidence catalogue.

22. Audit and assurance checklist.

Agentic AI operating emphasis

The training uses the four practical dimensions reflected in IMDA's current 2026 agentic-AI framework:

- assess and bound risks and capabilities upfront;
- make humans meaningfully accountable at significant checkpoints;
- implement technical controls and processes throughout the lifecycle;
- enable responsible end-user behavior through transparency and education.

The May 2026 update is reflected through added attention to multi-agent systems, third-party agents, case-based operationalization and automation-bias controls.

Stage-3 evidence model

Every framework concept is translated using:

****Source concept → organizational interpretation → risk/control objective → owner → implementation → evidence → test/assurance method → limitation/non-equivalence note****

Binding law, voluntary framework guidance, assurance tooling, implementation examples and proposed standardisation work remain clearly distinguished.

Publication gates

- [x] Official IMDA source verification refreshed at Stage 3 entry.
- [x] Current MGF/GenAI/AI Verify/Agentic materials reconciled for controlled drafting.
- [x] Controlled-English full training package drafted after Manual 47 advanced beyond Stage 3.
- [x] Cross-framework false-equivalence rules embedded.
- [x] Technical agentic-AI control architecture drafted.
- [] Accountable-human technical/governance and source-mapping review where genuinely required.
- [] es-419 and pt-BR controlled localization.
- [] Reproducible candidate build and six-binary artifact.
- [] Accessibility/rendered QA and provenance.
- [] Exact-byte staging and final publication reconciliation.

****Fail-closed:**** retained genuine-human substantive review evidence may not be fabricated. Under the rolling anti-halt rule, a Manual 48 blocker does not halt safe work on Manual 49 and later.

Manual 48 — Singapore AI Governance Training Modules

****Controlled English draft — Stage 3****

****Currentness baseline:**** 1 September 2026

****Purpose:**** convert Singapore's practical AI-governance ecosystem into an enterprise operating model without treating voluntary guidance or assurance tools as binding law or certification.

Module 1 — Singapore's practical AI-governance ecosystem

Learning objective

Distinguish the roles of the Model AI Governance Framework, the Generative AI framework, AI Verify, the Agentic AI framework and related assurance/testing initiatives.

Operating rule

For every source used in a control decision, label it as one of: governance guidance, assurance/testing framework, implementation example, proposed standardisation work, or binding law from another jurisdiction. Do not collapse those categories.

Evidence

- framework inventory;
- source/version register;
- applicability note;
- non-equivalence statement.

Module 2 — Foundational Model AI Governance Framework

Operationalise four enduring organisational themes:

- 23. internal governance structures and accountability;
- 24. appropriate human involvement in AI-augmented decisions;
- 25. operations management across the AI lifecycle; and
- 26. stakeholder interaction and communication.

Control pattern

****Policy** → accountable owner → risk assessment → lifecycle control → evidence → testing → remediation.******

Minimum evidence

- AI policy and governance charter;
- RACI / accountable executive and system owner;
- AI inventory and use-case classification;
- documented human-oversight model;
- data/model lifecycle procedures;
- stakeholder communication plan.

Module 3 — Human involvement and decision accountability

Training outcome

Participants must be able to justify the selected level of human involvement rather than merely state that a human is 'in the loop'.

Control questions

- What decision or action is the AI influencing?
- What harm could occur if the AI is wrong?
- Is the human able, trained and authorised to intervene?
- Is the review point early enough to prevent harm?
- Is the information presented to the reviewer sufficient for meaningful judgment?
- Is automation bias monitored?

Evidence

- decision-rights matrix;
- approval thresholds;
- override/stop procedure;
- reviewer training record;
- sampled approval logs;
- override and exception metrics.

Module 4 — Operations management and lifecycle controls

Establish controls for data quality, model/system validation, robustness, monitoring, tuning/change, incident response and retirement.

Required operating artifacts

- lifecycle control plan;
- data lineage and quality evidence;
- validation/test report;
- monitoring thresholds;
- change/revalidation triggers;
- incident and rollback procedure;
- retirement/decommission record.

Module 5 — Stakeholder interaction and communication

Train teams to communicate AI use clearly, provide understandable information, maintain feedback channels where appropriate and avoid claims that exceed evidence.

Evidence

- user notices;
- model/system limitations statement;
- feedback/complaint process;
- communications approval record;
- claims register.

Module 6 — Generative AI governance

Treat the GenAI framework as an ecosystem extension to foundational governance. Enterprise training focuses on the dimensions most directly actionable by deployers and developers:

- accountability and governance;
- data and provenance;
- trusted development and deployment;
- incident reporting and monitoring;
- testing and assurance;
- security and safety;
- content provenance and transparency;
- third-party/supply-chain responsibility; and
- societal/user impact considerations.

GenAI control questions

- What training, retrieval, prompt and tool data enter the system?
- How are sensitive data and confidential sources constrained?
- What evaluation set demonstrates acceptable performance?
- What hallucination, toxicity, security and privacy failure modes are tested?
- How are generated outputs identified or qualified where appropriate?
- How are model/provider changes detected and revalidated?

Module 7 — AI Verify assurance and testing

Training principle

AI Verify is used as an assurance/testing mechanism, not as proof that an AI system is universally safe, unbiased, compliant or certified.

Enterprise workflow

27. define the claims to be assessed;
28. identify applicable process checks and technical tests;
29. document data, test conditions and limitations;
30. execute reproducible tests where feasible;
31. review failures and residual risk;
32. record management disposition;
33. re-test after material change.

Evidence

- test plan;
- process-check responses;
- technical test outputs;
- reproducibility record;
- exception log;
- remediation plan;
- management sign-off.

Module 8 — Agentic AI: assess and bound risks upfront

Use the current Agentic AI framework as the primary operating reference for systems that can plan, select tools, interact with other agents or execute actions.

Required assessment dimensions

- intended use and prohibited use;
- autonomy level;
- tool/API access;
- data access and sensitivity;
- financial/operational authority;
- external communication capability;
- code execution capability;
- multi-agent interaction;
- third-party-agent dependency;
- reversibility of actions;
- potential blast radius.

Controls

- capability inventory;
- least-privilege access;
- action allowlist/denylist;
- transaction/value thresholds;
- environment segmentation;
- rate/volume limits;
- safe default on uncertainty.

Module 9 — Agentic AI: meaningful human accountability

Human approval must be tied to ****significant checkpoints**** rather than cosmetic review.

Examples of significant checkpoints

- release to production;
- access to a new high-impact tool or dataset;
- external payment or contractual action;
- privileged system change;
- irreversible deletion;
- safety/security exception;
- material model/toolchain change;
- escalation after abnormal behaviour.

Evidence

- checkpoint catalogue;
- approval policy;
- identity of approver;
- approval timestamp and context;
- rejected/overridden action log;
- escalation record.

Module 10 — Agentic AI: lifecycle technical and process controls

Baseline controls

- authenticated agent/service identity;
- least privilege and scoped credentials;
- tool allowlisting;
- input/output validation;
- prompt/tool boundary protection;
- sandboxing where appropriate;
- action logging and provenance;
- anomaly monitoring;
- rate limiting;
- kill switch / containment mechanism;
- dependency inventory;
- third-party-agent risk assessment;
- change control and revalidation.

Multi-agent emphasis

Document delegation chains, inter-agent permissions, shared-memory boundaries, conflict-resolution logic and how responsibility is reconstructed across agent interactions.

Module 11 — Agentic AI: end-user transparency and responsibility

Users should understand when they are interacting with an agent, the agent's role and limits, the kinds of actions it can take, what review is expected of the user and where to report problems.

Evidence

- agent identity disclosure;
- capability/limitation notice;
- user training;
- feedback channel;
- escalation instructions;
- sampled interaction review.

Module 12 — Third-party AI and agent supply chain

Control objectives

- know which provider/model/agent/tool is being used;
- understand material dependencies and data flows;
- contract for security, privacy, change notification and incident cooperation;
- monitor provider changes;
- revalidate after material model, API, policy or capability changes;
- prevent inherited credentials or excessive privileges.

Evidence

- vendor risk assessment;
- architecture/data-flow diagram;
- contract/control mapping;
- provider-change log;
- exit/continuity plan.

Module 13 — Automation bias and operational challenge

The May 2026 Agentic AI update specifically strengthens practical attention to automation bias.

Training exercises

- identify decisions where users are likely to defer to AI output;
- design counter-signals, independent checks or forced review;
- measure override rates and reviewer disagreement;
- investigate near-zero override rates where independent judgment is expected.

Module 14 — Assurance, evidence and auditability

Every major control must be testable.

Evidence rule

For each control maintain:

Source concept → organisational interpretation → control objective → control owner → implementation → evidence → test method → result → finding → remediation → residual risk.

Module 15 — Cross-framework mapping without false equivalence

Map Singapore guidance to Manual 46 universal controls and, where useful, to EU AI Act, ISO/IEC 42001 and NIST AI RMF control themes. Mapping is for operational reuse, not a statement that satisfying one framework automatically satisfies another.

Required mapping fields

- Singapore source concept;
- target framework concept;
- relationship type: direct / partial / supporting / contextual;
- important differences;
- additional evidence required;
- legal/standards caveat.

Module 16 — Enterprise implementation blueprint

30-day foundation

- establish AI governance owner and inventory;
- classify active AI/GenAI/agentic use cases;
- identify highest-risk agentic deployments;
- define significant human checkpoints;

- establish evidence repository.

60-day operating model

- implement lifecycle controls;
- perform vendor and tool-access reviews;
- establish testing/assurance cadence;
- formalise incident and containment procedures;
- train reviewers and end users.

90-day assurance cycle

- execute independent sampling/testing;
- validate high-risk agent/tool permissions;
- review automation-bias indicators;
- close findings;
- produce governance dashboard and management attestation.

Completion standard

A learner completes Manual 48 when they can take a Singapore governance concept and produce a defensible enterprise control, evidence set, test procedure and limitation statement without claiming legal equivalence or certification.

Manual 48 — Practical Scenarios

Scenario 1 — Customer-service GenAI with confidential data risk

A customer-service team deploys a GenAI assistant using retrieval over policy documents and selected customer records. The team proposes sending every conversation to the model provider for quality improvement.

Learner tasks

34. identify data-governance and confidentiality risks;
35. define permitted retrieval sources and data classes;
36. establish input/output filtering and retention controls;
37. create an evaluation plan for hallucination, leakage and harmful responses;
38. define the evidence needed for assurance.

Expected control outcome

Use data minimisation, source allowlisting, contractual/provider controls, logging, evaluation, human escalation and incident response. Do not infer that use of AI Verify alone proves the system safe or legally compliant.

Scenario 2 — Finance agent with payment authority

An agent receives invoices, matches them to purchase orders and can initiate payment instructions up to a configurable threshold.

Learner tasks

- determine whether the use case is suitable for agentic automation;
- bound the agent's powers;
- define significant human checkpoints;
- specify identity, credential and tool-access controls;
- design action provenance and a kill switch.

Minimum safeguards

- separate agent identity;
- least privilege;
- payment/value limits;
- allowlisted counterparties or payment rails;
- mandatory approval for exceptions/high-value transactions;
- complete transaction/action log;
- independent reconciliation;
- rapid disablement capability.

Scenario 3 — HR recruiting agent and automation bias

A recruiting agent ranks candidates, drafts screening recommendations and schedules interviews. Recruiters rarely override its ranking.

Learner tasks

- identify where automation bias may occur;
- decide what human review is meaningful;
- design monitoring that detects excessive deference;
- define candidate-facing transparency and escalation where appropriate;
- document limitations and prohibited uses.

Evidence

- reviewer guidance;
- sampled override/disagreement data;
- fairness/performance testing where applicable;
- decision rationale;
- issue-remediation log.

Scenario 4 — Third-party coding agent

A software team connects a third-party coding agent to source repositories, issue trackers, CI/CD and cloud development environments.

Learner tasks

39. inventory tools and credentials;
40. classify permitted and prohibited actions;
41. determine whether production deployment or privileged changes require human approval;
42. design sandboxing, branch protection and secrets controls;
43. define provider-change and incident-response requirements.

Expected control outcome

Separate identities, short-lived/scoped credentials, protected branches, code review, CI security checks, restricted production access, immutable logs and revocation/containment procedures.

Scenario 5 — Multi-agent travel operations

A travel platform uses one agent to interpret customer intent, another to search inventory, another to create bookings, and another to handle disruption rebooking.

Learner tasks

- map delegation paths;
- define which agent can instruct which other agent;
- prevent confused-deputy and privilege-escalation patterns;
- define user confirmation requirements for irreversible or costly actions;
- preserve end-to-end action provenance.

Evidence

- inter-agent trust-boundary diagram;
- permissions matrix;
- delegation/event logs;
- replayable transaction trail;
- exception and rollback record.

Scenario 6 — Public-facing advisory chatbot

An organisation deploys a chatbot that explains eligibility criteria and suggests next steps but is not authorised to make formal eligibility decisions.

Learner tasks

- communicate the agent's actual authority;
- prevent the system from presenting guidance as a binding determination;
- establish escalation to human staff;
- test for hallucinated policy claims;
- maintain source/version control.

Scenario 7 — AI Verify assurance exercise

A project team says, 'The system passed AI Verify, therefore it is compliant and safe.'

Learner task

Rewrite the conclusion into an evidence-based assurance statement.

Expected answer pattern

State which claims, process checks and technical tests were assessed; identify the test conditions, results, limitations and residual risk; explicitly avoid certification/compliance claims not supported by the evidence.

Scenario 8 — Agent capability expansion after vendor update

A provider update adds web browsing, file upload and autonomous tool selection to a previously bounded agent.

Learner tasks

- identify the change as a revalidation trigger;
- reassess blast radius and data/tool access;
- update human checkpoints;
- re-run relevant safety/security tests;
- decide whether deployment should remain suspended until controls are updated.

Scenario 9 — Executive governance review

A governance committee receives a dashboard showing 30 AI systems, five GenAI systems and three agents. Two agents have not completed current access reviews, one has no tested kill switch, and one vendor changed models without prior notice.

Learner tasks

Prioritise findings, assign owners, define remediation deadlines and specify which systems should be conditionally restricted or suspended.

Scenario 10 — Cross-framework mapping challenge

A business asks whether implementing Singapore's Agentic AI framework means the organisation 'complies with the EU AI Act and ISO/IEC 42001.'

Required learner response

Reject automatic equivalence. Produce a mapping that identifies supporting overlaps, partial relationships, differences in legal/standards status and additional evidence or obligations required by each target regime.

Manual 48 — Control and Evidence Workbook

Use this workbook to translate Singapore AI-governance guidance into auditable enterprise controls. Each row should be completed for the organisation's actual AI system and retained with evidence.

| ID | Control objective | Implementation expectation | Minimum evidence | Test / assurance method | Common failure |

|---|---|---|---|---|---|

| SG-AI-01 | Governance accountability | Named accountable executive, system owner and control owners | charter, RACI, committee terms | verify ownership and sampled decisions | ownership exists only on paper |

| SG-AI-02 | AI inventory | Maintain current inventory of AI, GenAI and agents | inventory, owner, purpose, provider, status | reconcile against procurement/cloud/repos | shadow AI omitted |

| SG-AI-03 | Use-case risk assessment | Assess impact, data, autonomy, users and failure modes before deployment | risk assessment, disposition | sample completeness and approvals | generic risk template with no use-case analysis |

| SG-AI-04 | Human involvement | Define meaningful review/approval where harm or irreversible action warrants it | decision-rights matrix, approval logs | test whether reviewer can intervene | rubber-stamp human in the loop |

| SG-AI-05 | Data governance | Control collection, retrieval, sensitive data, lineage and retention | data-flow map, source list, retention rules | trace sampled data paths | unrestricted RAG/data sources |

| SG-AI-06 | Lifecycle operations | Define validation, monitoring, change, incident and retirement controls | lifecycle plan, test reports, monitoring | inspect material changes and revalidation | model/provider changes not detected |

| SG-AI-07 | Stakeholder transparency | Communicate AI use, role and limitations accurately | notices, user guidance, claims register | compare claims to evidence | overstated capability or compliance claims |

| SG-GEN-01 | GenAI evaluation | Evaluate hallucination, harmful output, security/privacy and task performance | evaluation plan/results | reproduce selected tests | only benchmark accuracy measured |

| SG-GEN-02 | Prompt/RAG boundary | Bound prompt, retrieval and tool inputs | source allowlist, filters, prompt controls | adversarial retrieval/prompt tests | confidential or poisoned sources admitted |

| SG-GEN-03 | Content provenance | Identify or qualify generated content where relevant | metadata/notice policy | sample outputs | no traceability for high-impact content |

| SG-ASSURE-01 | Assurance claims | Tie every assurance statement to actual checks/tests and limitations | test scope, report, limitation note | trace claims to evidence | 'passed AI Verify = compliant' |

| SG-ASSURE-02 | Reproducible testing | Preserve test conditions, data/version and results | test config, versions, logs | rerun selected tests | results cannot be reproduced |

| SG-AG-01 | Agent capability inventory | Record autonomy, tools, data, actions and external communication | capability matrix | compare with live permissions | undocumented tool capability |

| SG-AG-02 | Bound agent powers | Apply least privilege, allowlists, limits and safe defaults | IAM policy, tool policy, thresholds | attempt prohibited action | broad inherited credentials |

| SG-AG-03 | Significant human checkpoints | Require approval at high-impact/irreversible boundaries | checkpoint catalogue, approval records | sample transactions | approval occurs after action |

| SG-AG-04 | Agent identity | Give agents/services attributable identities | identity records, auth config | trace sampled action to identity | shared human/service account |

| SG-AG-05 | Action provenance | Record requests, plans, tool calls, results and material actions | immutable/event logs | reconstruct sampled event | fragmented or missing logs |

| SG-AG-06 | Containment | Provide tested disable/kill/credential-revocation mechanisms | runbook, test evidence | tabletop or controlled test | kill switch exists but untested |

| SG-AG-07 | Multi-agent boundaries | Define delegation, trust and shared-memory boundaries | architecture diagram, permissions | trace cross-agent action | implicit privilege propagation |

| SG-AG-08 | Third-party agent risk | Assess external agent/provider changes and dependencies | vendor assessment, contract, change log | review update notification/revalidation | silent capability expansion |

| SG-AG-09 | Automation-bias control | Preserve independent human judgment where required | reviewer training, override metrics | analyze override/disagreement rates | near-zero overrides ignored |

| SG-AG-10 | End-user responsibility | Disclose agent identity, role, limits and escalation | user notice/training | sample user journeys | user believes agent has authority it lacks |

| SG-AG-11 | Change/revalidation | Define material changes that trigger new assessment/testing | trigger list, change tickets | sample model/tool/provider updates | no revalidation after capability change |

| SG-AG-12 | Incident response | Detect, contain, investigate and remediate AI/agent incidents | IR playbook, incident records | tabletop/sample incident | AI incidents handled outside IR process |

| SG-XW-01 | Cross-framework non-equivalence | Map overlaps without claiming automatic legal/standards equivalence | crosswalk, caveat column | review mappings | framework alignment misrepresented as compliance |

Evidence register template

For every implemented control record:

- Control ID
- AI system / use case
- Accountable owner
- Control operator
- Evidence location
- Evidence period / timestamp
- Model/provider/tool versions
- Test procedure
- Test result
- Exception / finding
- Remediation owner and due date
- Residual risk decision
- Revalidation date

Agent action review sample

For a sample of material agent actions, verify:

44. attributable agent/service identity;

45. authenticated user or upstream initiator where applicable;
46. relevant instruction/context;
47. selected tool/API;
48. permission decision;
49. human checkpoint if required;
50. action result;
51. error/exception handling;
52. downstream consequences;
53. rollback/containment availability.

Management dashboard minimums

- AI systems by risk tier;
- GenAI/agent systems by business owner;
- overdue risk assessments;
- overdue access reviews;
- high-impact actions requiring approval;
- approval override/rejection rates;
- unresolved assurance findings;
- provider/model changes awaiting revalidation;
- AI/agent incidents and near misses;
- tested/untested containment mechanisms;
- third-party-agent dependencies;
- overdue remediation items.

Release caveat

This workbook operationalises Singapore governance guidance. Completion of the workbook is not certification and does not by itself establish compliance with Singapore law, the EU AI Act, ISO/IEC 42001, NIST AI RMF, or any other external requirement.